

DZHUSALY, KazakhstanWMO: **359530**Lat: **45.510N**Lon: **64.090E**Elev: **338**StdP: **14.52**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-8.2	-18.6	2.0	-12.7	-13.4	2.7	-7.7	47.2	5.8	39.0	11.1	8.8	50	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	102.0	67.2	99.0	66.0	96.2	65.2	70.1	94.0	68.3	92.4	66.9	91.0	11.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	87.3	77.5	59.7	77.5	77.7	56.9	70.0	78.2	34.3	93.8	32.8	92.6	31.7	90.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 34.1	28.9	25.8	DB	-16.6	107.4	7.2	2.7	-21.8	109.3	-26.0	110.9	-30.1	112.4	-35.3	114.3	(4)
(5)			WB	-16.6	73.0	7.1	3.5	-21.7	75.5	-25.9	77.5	-29.9	79.4	-35.0	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	15.7	20.4	38.8	55.3	68.1	78.0	81.3	78.7	65.6	49.3	33.8	21.2		
	DBStd	25.06	12.26	13.73	10.67	9.48	7.21	6.73	5.99	6.86	8.40	9.05	10.37	10.93		
	HDD50	3833	1064	829	369	57	1	0	0	0	3	125	491	894		
	HDD65	6835	1529	1249	814	309	51	4	0	3	91	491	935	1359		
	CDD50	4073	0	0	21	215	561	840	969	888	471	103	5	0		
	CDD65	1599	0	0	0	18	146	393	504	426	109	3	0	0		
	CDH74	22110	0	0	16	308	1984	5512	7196	5664	1345	85	0	0		
	CDH80	11886	0	0	4	83	823	3053	4172	3179	554	18	0	0		
(14) Wind	WSAvg	10.2	10.3	10.3	11.3	11.4	11.2	10.6	10.1	10.2	9.5	8.9	8.8	9.9		
(15) Precipitation	PrecAvg	5.4	0.6	0.5	0.6	0.6	0.6	0.4	0.2	0.2	0.1	0.4	0.6	0.6		
	PrecMax	8.1	1.3	1.1	1.5	1.7	1.6	1.7	2.1	1.6	0.9	2.2	1.4	1.7		
	PrecMin	2.9	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	1.4	0.3	0.3	0.3	0.5	0.4	0.4	0.4	0.3	0.2	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	52.0	74.7	87.8	96.2	103.9	107.2	104.2	97.5	82.0	64.6	44.7		
		MCWB	39.2	45.7	54.8	60.6	63.3	67.2	71.1	66.8	64.0	57.3	52.7	39.4		
	2%	DB	36.8	44.0	65.6	81.8	91.9	100.3	102.7	100.6	91.1	75.5	57.2	40.1		
		MCWB	35.2	38.9	51.0	58.2	62.0	66.1	68.0	66.0	60.8	53.8	47.9	37.4		
	5%	DB	34.1	39.3	59.7	76.9	88.5	97.1	99.2	97.7	86.2	70.4	52.3	36.8		
		MCWB	32.7	35.9	47.8	55.4	60.7	64.9	66.8	65.2	59.6	51.9	45.1	34.8		
	10%	DB	31.9	35.9	54.4	72.3	84.8	94.0	96.1	94.5	81.6	65.4	47.7	34.0		
		MCWB	30.6	33.3	45.4	53.5	59.7	63.9	66.2	64.5	57.7	50.0	42.1	32.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	46.2	56.0	63.2	68.2	71.5	74.7	71.7	65.5	58.7	54.8	41.5		
		MCDB	41.7	51.6	72.4	81.4	88.6	94.3	94.8	91.9	91.7	77.9	61.5	43.4		
	2%	WB	35.6	39.7	52.0	60.3	64.5	68.1	70.8	69.0	62.7	55.5	49.3	37.8		
		MCDB	36.6	42.8	62.6	76.2	84.4	92.2	95.6	93.5	87.0	71.0	55.6	39.8		
	5%	WB	32.9	36.0	49.0	57.4	62.6	66.7	69.0	67.1	60.6	53.0	45.4	35.0		
		MCDB	34.0	39.0	58.4	73.3	83.1	90.9	92.9	92.5	83.3	67.9	51.1	36.5		
	10%	WB	30.6	33.5	46.1	54.8	60.8	65.4	67.5	65.4	58.5	50.7	42.5	32.5		
		MCDB	31.9	35.6	53.8	68.8	81.4	89.6	91.1	90.9	79.9	63.9	47.4	34.0		
(35) Mean Daily Temperature Range	MDBR		13.7	15.9	19.6	23.2	25.0	25.5	24.4	25.0	25.1	23.6	16.2	12.7		
	5% DB	MCDBR	11.4	17.8	28.0	29.7	29.7	28.9	27.5	28.6	29.7	30.4	23.4	13.4		
		MCWBR	10.3	13.4	15.8	13.4	11.5	10.3	9.9	10.3	12.3	15.7	15.5	11.0		
	5% WB	MCDBR	10.6	16.4	26.0	26.4	26.5	26.1	24.8	25.5	27.6	27.8	20.8	12.0		
		MCWBR	9.7	12.7	15.8	13.5	12.0	10.5	10.3	10.4	12.5	15.4	14.4	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.367	0.428	0.469	0.471	0.450	0.432	0.417	0.383	0.387	0.370	0.349		
	taud		2.074	2.041	1.987	1.920	1.969	2.085	2.162	2.180	2.220	2.124	2.126	2.090		
	Ebn at Noon		228	247	248	250	255	261	264	263	262	238	216	209		
	Edh at Noon		32	40	49	57	56	50	46	43	38	36	30	28		
(44) All-Sky Solar Radiation	RadAvg		436	795	1207	1699	2163	2337	2277	2062	1613	1016	566	358		
	RadStd		91	116	98	120	121	115	98	76	51	61	55	59		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-3.49	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+1.58	N/A	-1.51	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page